2026 H2 Prelim Paper 2 Solutions

1 (a) Systematic errors will result in all readings being either above or below the accepted value. B1

Random errors will result in readings being scattered around the accepted value (ie equal probability being above or below the accepted value)

Systematic errors cannot be eliminated by repeating readings and then averaging. B1

Random errors can be reduced by repeating the readings and averaging.

Systematic error: same magnitude and sign

Random error: different magnitude and sign

(b)(i) $T = 2\pi\sqrt{(\frac{0.50}{9.8})} = 1.419$ A1

$\frac{\Delta T}{T} = \frac{1}{2}(\frac{\Delta L}{L} + \frac{\Delta g}{g})$ M1

$\Rightarrow \Delta T = \frac{1}{2}(\frac{0.2}{50.0} + \frac{0.1}{9.8}) \times 1.419 = 0.01$ (to 1 s.f) A1

T = 1.42 ± 0.01 s

(b)(ii) Due to human reaction time, large number of oscillations will reduce the fractional or percentage uncertainty of the calculated value of T. B1

2 (a) The gradient of the graph gives acceleration. B1

Acceleration is largest initially at 9.81 m s^{-2} and decreases to zero at 1.5 s B1

(b) Drawing of a tangent at t = 0.25 s C1

a = 1.3 m s $^{-2}$ (Allow answer in the range 1.2 to 1.4 m s^{-2}) A1

(c) $F_{net} = W - D$

$\Rightarrow 1.2 \times 10^{-2} \times 1.3 = 1.2 \times 10^{-2} \times 9.81 - D$ C1

D = 0.100 N A1

(d) (i) Drag force is the same since speed is the same. B1

Weight is greater as total mass of sand-filled ball is greater, thus the net force will be larger. B1

(d)(ii) At terminal speed, weight = Drag

Since weight now is greater, drag force will be higher. C1

As drag force increases with speed, the terminal speed reached will be higher

A1

3 (a) It has a constant vertical acceleration and a constant horizontal velocity. B1

(b)(i) Vertical component of velocity at P = $8.2 \tan 60^{o} = 14.2$ m s^{-1} C1

Using $v = u + a\,t$

$14.2 = 0 - 9.81\,t$

$\Rightarrow \; t = 1.45$ s A1

(b)(ii) $x = u_x t = 8.2 \times 1.45 = 11.9$ m A1

(c) Displacement is the shortest distance (or straight-line distance) from top to point P, C1

so it is less than the distance along the curved path A1

(d)(i) Smooth curved path above given path, and A1

Hit ground at a more acute angle A1/2

(d) (ii) Smooth curved path below given path, and A1

Hit ground at steeper angle A1/2

4 (a) The gravitational potential at a point is the work done per unit mass by B1

an external agent in bringing a point mass from infinity to that point B1

without a change in kinetic energy.

(b) $\phi = \phi_M + \phi_M$

$= -\dfrac{GM}{r} + \left(-\dfrac{GM}{r}\right)$ C1

$= -\dfrac{2GM}{r}$

$= -\dfrac{2(6.67 \times 10^{-11})(3.5 \times 10^{30})}{\frac{2.0 \times 10^{11}}{2}}$

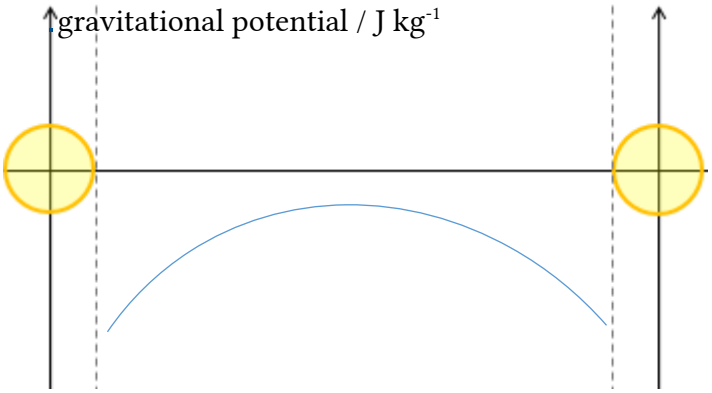
$= -4.67 \times 10^{9} = -4.7 \times 10^{9}$ J kg^{-1} A1

(c) $KE_i + PE_i = KE_f + PE_f$

$\frac{1}{2}mv^2 + m(-4.67 \times 10^{9}) = 0 + 0$ C1

$v = 9.66 \times 10^{4}$ m s^{-1} A1

(d)



Correct shape, symmetrical — B1
Below zero — B1

(e) Gravitational force provides for centripetal force

$$F_G = F_c$$

$$\frac{GMM}{d^2} = MR\omega^2 \qquad \text{C1}$$

$$\frac{6.67\times10^{-11}(3.5\times10^{30})}{(2.0\times10^{11})^2} = (1.0\times10^{11})\left(\frac{2\pi}{T}\right)^2$$

$$T = 2.6\times10^{7}\ \text{s} \qquad \text{A1}$$

5a	$+\Delta U$ increase in internal energy $+q$ heat energy transferred to the system $+w$ work done on system (1 error -1 mark)	B2
bi	$W = p\Delta V = 5.2\times10^{5}\times(5.0-1.6)\times10^{-4} = 177\ \text{J}\ \Delta U = q + w = 442 - 177\ \Delta U = 265\ \text{J}$	M1 M1
bii	No molecular/microscopic potential energy. Internal energy decreases, so total molecular/random/microscopic kinetic energy decreases. Mean molecular kinetic energy is proportional to thermodynamic temperature and decreases, so temperature decreases.	B1 B1 B1
biii	$\Delta U + 265 - 313 = 0\ \Delta U = 48\ \text{J}$	A1
biv	$pV = NkT\ 5.2\times10^{5}\times1.6\times10^{-4} = N\times1.38\times10^{-23}\times(273+227)\ N = 1.2\times10^{22}$	C1 A1

6 (a) (i) When the white light is incident on the gas, photons with wavelengths that correspond to the energy difference between any two energy levels are absorbed by the atoms. During de-excitation, these photons are emitted in all directions. In the resulting spectrum, these wavelengths correspond to dark spectral lines on a coloured background. [B1] [B1] [B1]

(ii)

$$\Delta E_{12} = hf = h\frac{c}{\lambda_{12}}$$

$$\lambda_{12} = \frac{hc}{\Delta E_{12}}$$

$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{\left[(-3.04) - (-5.14)\right] \times (1.60 \times 10^{-19})}$$ [M1]

$$= 592 \text{ nm (shown)}$$ [A0]

(iii)

$$\lambda_{13} = \frac{hc}{\Delta E_{13}}$$

$$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{\left[(-1.53) - (-5.14)\right] \times (1.60 \times 10^{-19})}$$

$$= 344 \text{ nm}$$ [M1]

Since wavelength of light corresponding to transition from E_1 to E_3 is already outside visible region, there is no need to test for transition from E_1 to E_4.

Hence, only one absorption spectral line, corresponding to the E_1 to E_2 transition, is observed. [A1]

OR OR

$$E_{max} = \frac{hc}{\lambda_{min}} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{400 \times 10^{-9}} = 4.97 \times 10^{-19} \text{ J} = 3.108 \text{ eV}$$

$$-5.14 + 3.108 = -2.03 \text{ eV} < -1.53 \text{ eV}$$ [M1]

Since only level E_2 is attainable from level E_1, only one absorption spectral line, corresponding to the E_1 to E_2 transition, is observed. [A1]

(b) (i)

$$\text{KE} = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m}$$

$$= \frac{(6.63 \times 10^{-34} / 592 \times 10^{-9})^2}{2 \times (9.11 \times 10^{-31})}$$ [M1]

$$= 6.88 \times 10^{-25} \text{ J}$$

$$= \frac{6.88 \times 10^{-25}}{1.6 \times 10^{-19}} = 4.30 \times 10^{-6} \text{ eV (3 s.f.)}$$ [A1]

(ii)
- Higher resolution / higher magnification B1
- due to shorter wavelengths used as energy of the electron can be increased by acceleration it to high speeds. B1

c **i** Probability density of finding the electron at distance r. B1

ii Probability density is greatest near $r = 0$. B1

Therefore the electron is most likely to be detected close to the nucleus. B1

7a	splitting up of a (large) nucleus / nuclide after absorbing a neutron	B1
bi	(neutrons) one of the products prompt further fission or neutrons collide with further nuclei	B1
bii	The number of neutrons emitted will be very large as the stages proceed and hence leading to a very large number of reactions	B1
biii	92 – 36 = 56	A1
biv	The range of percentage yield varies over a wide range of magnitude and hence in order to show the variation of the values in a single graph, a logarithmic scale is used.	B1
	Percentage yield of having 99 = 6 % Percentage yield of masses equal = 1.5 x 10^{-2} % Both values to get 1 mark each Therefore, the ratio = 6/(1.5 x 10^{-2}) = 400	B1 B1
	Gamma photon	B1
	1.00kg / 235 u = 1.00 kg / 235 x (1.66 x 10^{-27} kg) =2.56 x 10^{24}	C1 A1
	Energy of fission products = 2.5634 x 10^{24} x 167 x 10^{6} x 1.6 x 10^{-19} Electrical energy = 0.25 x 2.5634 x 10^{24} x 167 x 10^{6} x 1.6 x 10^{-19} = 1.71 x 10^{13} J	C1 C1 A1
	Power = energy / time = 1.7124 x 10^{13} / (24 x 60 x 60) = 1.98 x 10^{8} W	 C1 A1
	$^{99}_{42}Mo$ and its decay products have a very long half life. This means that the storage of $^{99}_{42}Mo$ will have to be in a lead container and put away for a very long time. (Max 2 marks for the next 3 points) On the other hand, $^{140}_{54}Xe$ has a very short half life and many decay products. For the same amount of these fission products, $^{140}_{54}Xe$ having many more fission products imply a greater safety concern for 2 reasons: 1. the total activity of $^{140}_{54}Xe$ and its daughter products is much higher than $^{99}_{42}Mo$. 2. with a short half-life, it means the decay constant is much greater, using	B1 B1 B1

	$A=\lambda N$, the activity from $^{140}_{54}Xe$ is much higher. The much higher activity implies that $^{140}_{54}Xe$ is very hazardous immediately after it is produced, and must be handled with great care.	